

Probability

Probability

- Probability is the basic language of statistics
- Probability vs Statistics
 - Probability: “a fair coin is tossed 10 time; what is the probability of getting 8 heads?”
 - Statistics: “a coin is tossed 10 times, showing 8 heads; is the coin fair?”
- Origins of probability: card/dice games, gambling

Experiments, Outcomes, and Sample Spaces

- An *experiment* governed by chance has several possible *outcomes*
- The collection of all possible outcomes in an experiment is the *sample space*
- Examples
 - Tossing a coin
 - Rolling a die
 - Rolling two dice

Events

- *Event* = a collection of possible outcomes in an experiment
 - A selected card is an Ace
 - A roll of a die is 5 or higher
- Usually denoted by capital letters
- $P(A)$ = Probability of event A

Properties of Probability

- The probability of an event A , $P(A)$, is a number between 0 and 1 that measures the likelihood that A will occur.
- $0 \leq P(A) \leq 1$ always
- $P(A) = 0$ means that A is *impossible (discrete events)*
- $P(A) = 1$ means that A *always happens (discrete events)*

Mutually Exclusive Events

- Two events are said to be mutually exclusive if, when one of the two events occur in an experiment, the other cannot.
- For example, if a fair coin is tossed once, the events T (Tails) and H (Heads) are mutually exclusive.

Mutually Exclusive Events

- Example

- Let A be the event of being dealt an Ace.
- Let B be the event of being dealt a spade.
- Let C be the event of being dealt a heart.
- Let D be the event of being dealt a club.
- Let E be the event of being dealt a diamond.
- Then events B,C,D, and E are mutually exclusive.
- But the events, A and B, A and C, ... are not mutually exclusive.

Simple Events

- Simple events are mutually exclusive events that represent the most basic outcomes of an experiment.
- The sample space of an experiment is the collection of all of its simple events.
- For example, if two fair coins are tossed then what are the simple events and thus what is the sample space?

Intersections and Unions

- **Intersection**

- A and B occur simultaneously
- When both events A and B occur, we refer to this as the intersection of A and B denoted by $A \cap B$.

- **Union**

- Either A or B (including the case where both A and B occur simultaneously)
- When either A or B or both events occur we refer to this as the union of A and B denoted by $A \cup B$

Probability Rule #1

- **The additive rule for mutually exclusive events**
- If two events A and B are mutually exclusive , then the probability that either A or B occurs is equal to the sum of their respective probabilities:
- $P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$

Probability Rule #2

- Suppose an experiment can result in one and only one of M equally likely mutually exclusive events and that m of these events result in event A . Then the probability of event A is $P(A) = \frac{m}{M}$

Probability Rule #3

- If we list all possible simple events (sample points) associated with an experiment, then the sum of their probabilities is equal to 1. In other words, the probability of the entire sample space is always 1.

Probability Rule #4

- The complement of an event , denoted by $\bar{A}$ or A^c , is the event that A does not occur.
- $P(A) = 1 - P(\bar{A})$
- The complement of $\bar{A}$ is the event A.

Conditional Probability

- What is the difference between the following two problems?
 1. What is the probability that the sum of two dice is 5?
 2. Given that the first die is less than 3, what is the probability that the sum of the two dice is equal to 5?

Conditional Probability

- The probability of an event A, given that an event B has (hypothetically) occurred, is called the conditional probability of A given B and is denoted $P(A|B)$

If the probability of one event does not depend on whether a second event has occurred, then the events are said to be independent.

- **Definition:** Two events A and B are said to be independent if $P(A|B) = P(A)$ and $P(B|A) = P(B)$

$P(A \cap B)$

Probability Rule #5

- If two events A and B are independent, then the probability that both A and B occur is

$$P(A \cap B) = P(A)P(B)$$

Probability Rule #6

Multiplicative Law of Probability

The probability that events A and B both occur is

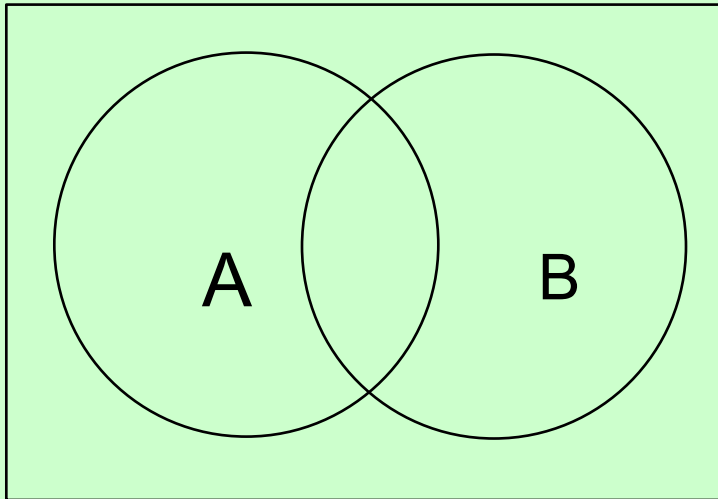
$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

$P(A \cup B)$

Probability Rule #7

Additive Law of Probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Example 1

We toss two coins and observe the upper faces of the coins.

- A: Observe at least one head
 - B: Observe at least one tail
1. What is the probability that both A and B occur?
 2. What is the probability that A or B or both events occur?

Example with Sample Space

- You roll two fair dice.
- Let A be the event the sum of two dice is 5.
- Let B be the event the first die is less than 3.
- Let C be the event the sum of two dice is 7.

1. $P(A)$

2. $P(B)$

3. $P(C)$

4. $P(A|B)$

8. $P(C|B)$

5. $P(A \cap B)$

9. $P(C \cap B)$

6. $P(A \cup B)$

10. $P(C \cup B)$

7. $P(B|A)$

11. $P(B|C)$

Example 3

Consider two events A and B , where $P(A)=.4$ and $P(B) =.2$.

1. If A and B are independent, what is, $P(A|B)$, $P(A \cap B)$, $P(A \cup B)$?

2. If $P(A|B) = .6$, then are A and B independent?
Find $P(A \cap B)$ and $P(B|A)$.

Example 4

- Choose at random a student from a class of 62 students

	USA	Europe	Latin America	Asia	Other	Total
Male	22	5	4	6	3	40
Female	13	3	1	3	2	22
Total	35	8	5	9	5	62

- Find
 - $P(\text{European}) = ?$
 - $P(\text{European and Female}) = ?$
 - $P(\text{Male or non-USA}) = ?$
 - $P(\text{Asian} \mid \text{Female}) = ?$
 - $P(\text{Male} \mid \text{not European}) = ?$

Baye's Theorem

- $$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|\bar{B})P(\bar{B})}$$

Denominator: $P(A) = P(A|B)P(B) + P(A|\bar{B})P(\bar{B})$

Example: Test for Disease

Conditional Probability – Example

- Jar A contains 7 red and 3 green marbles; jar B contains 15 red and 30 green. Flip a fair coin, and select a ball from jar A if tossed heads, or from jar B if tossed tails. Find
 1. $P(\text{red} \mid \text{heads}) = ?$
 2. $P(\text{red} \mid \text{tails}) = ?$
 3. $P(\text{red and heads}) = ?$
 4. $P(\text{red and tails}) = ?$
 5. $P(\text{red}) = ?$
 6. $P(\text{tails} \mid \text{green}) = ?$

Factorials and Permutations

- In how many ways can we place 4 people on a bench?
 $4 \cdot 3 \cdot 2 \cdot 1 = 24$
- “x factorial” = $x! = x(x - 1)(x - 2) \dots 1$
- The number of ways we can order x (distinct) elements in a row is $x!$
- By definition: $0! = 1$
- Permutations Example: There are 10 students running for president, vice president, and treasurer of a new club. How many different leadership teams exist?

Examples

- There are 5 messages in your voicemail, from Adams, Berger, Carlton, Davis, and Edgar, left in a random order.
 1. What is the probability that they are in alphabetical order (ABCDE)?
 2. What is the probability that Carlton is last?
 3. What is the probability that Adams is immediately after Davis?

Combinations

- In how many ways can we select a committee of 3 people from a group of 8?

- Answer is: $\frac{8!}{3!(8-3)!} = 56$

- The number of *combinations* to choose k elements from a set of n (when order doesn't matter) is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Combinations – Example

- An investor is considering investing in 5 firms, A, B, C, D, and E.
 1. If she chose two firms at random, what is the probability that these are A and B?
 2. If she chose three firms at random, what is the probability that E is among them?